

Non-Invasive Vitamin D Level Prediction via Machine Learning Analysis of Complete Blood Count Parameters

Likhitha Gopalasetti
Dept. of CSE, GST
GITAM Deemed to be University
Visakhapatnam, India
likhithagopalasetti@gmail.com

Nishitha Kunaparaju
Dept. of CSE, GST
GITAM Deemed to be University
Visakhapatnam, India
nishithakunaparaju@gmail.com

Chaitanya Sai Chintala
Dept. of CSE, GST
GITAM Deemed to be University
Visakhapatnam, India
chintalachaitanya2004@gmail.com

Tejaswi Kattamuri
Dept. of CSE, GST
GITAM Deemed to be University
Visakhapatnam, India
tejaswikattamuri3@gmail.com

Rita Roy
Dept. of CSE, GST
GITAM Deemed to be University
Visakhapatnam, India
ritaroy234@gmail.com

Abstract—Vitamin D deficiency is associated with critical health risks such as osteoporosis, immune disorders, and cardiovascular diseases. Current diagnostic methods rely on invasive and costly serum tests, necessitating non-invasive alternatives. This study proposes a machine learning (ML)-based approach to predict Vitamin D levels using routinely collected Complete Blood Count (CBC) parameters. We evaluate multiple ML algorithms, including decision trees, support vector machines (SVM), neural networks, and ensemble techniques (XGBoost, LightGBM, CatBoost), trained on preprocessed CBC datasets. Preprocessing involves feature selection, normalization, and missing value imputation to ensure robustness. Performance evaluation using mean absolute error (MAE), root mean squared error (RMSE), and R^2 metrics demonstrates that boosting algorithms outperform traditional models, with CatBoost and XGBoost achieving the highest accuracy. Key CBC parameters such as red blood cell count, mean corpuscular volume, and white blood cell distribution exhibit strong correlations with Vitamin D levels. This method enables non-invasive, cost-effective screening, facilitating early deficiency detection and personalized interventions. By leveraging existing CBC data, the approach reduces reliance on specialized tests, offering a scalable solution for public health systems.

Index Terms—Vitamin D deficiency, Machine Learning, Complete Blood Count, Non-Invasive Screening, CatBoost, XGBoost.

I. INTRODUCTION

Vitamin D, a fat-soluble secosteroid, is critical for calcium-phosphate homeostasis, bone mineralization, and immune modulation, with emerging roles in cardiovascular and neurological health [1]. Deficiency in this vitamin correlates with osteoporosis, autoimmune disorders, infections, and metabolic syndromes such as diabetes, underscoring its systemic importance [2]. Globally, deficiency prevalence remains high due to factors like limited sunlight exposure, dietary insufficiencies,

and demographic variables (e.g., age, skin pigmentation), necessitating widespread screening [3].

Current diagnosis relies on serum 25-hydroxyvitamin D [25(OH)D] quantification, the clinical gold standard. However, this approach is invasive, costly, and requires laboratory facilities, limiting routine use—especially in resource-limited settings [4]. Thus, non-invasive alternatives using existing clinical data are urgently needed.

This study investigates machine learning (ML) for predicting Vitamin D levels using Complete Blood Count (CBC) parameters—a routine, low-cost hematological test. CBC provides 20+ biomarkers, including erythrocyte/leukocyte indices and platelet metrics, which may indirectly reflect Vitamin D status through hematological interactions [5]. Recent advances in ML, particularly ensemble methods (XGBoost, CatBoost) and neural networks, enable robust pattern recognition in medical datasets [6]. We hypothesize that CBC-derived features, preprocessed via normalization and missing-data imputation, can train accurate regression models to estimate 25(OH)D concentrations.

Our contributions include:

- **Methodological Framework:** A comparative analysis of ML algorithms (decision trees, SVMs, gradient boosting, ANNs) for Vitamin D prediction, evaluated using MAE, RMSE, and R^2 .
- **Clinical Utility:** Identification of CBC biomarkers (e.g., mean corpuscular volume, leukocyte distribution) strongly associated with Vitamin D levels.
- **Resource Efficiency:** A pathway to repurpose CBC data for dual-use screening, eliminating dedicated Vitamin D tests.

By aligning with preventive healthcare paradigms, this approach could enable early deficiency detection, personalized

interventions, and reduced healthcare costs.

II. LITERATURE REVIEW

A. Vitamin D and Health Implications

Vitamin D's role in calcium homeostasis, bone health, and immune modulation is well-established. Studies by Holick et al. [1] highlight its critical function in preventing osteoporosis and autoimmune disorders. Recent epidemiological work [2] underscores its association with cardiovascular diseases and metabolic syndromes, emphasizing the global prevalence of deficiency, particularly in regions with limited sunlight exposure.

B. Traditional Diagnostic Methods

Serum 25(OH)D quantification via ELISA, HPLC, and LC-MS/MS remains the diagnostic gold standard [3]. While these methods offer high accuracy, their invasiveness, cost (\$50–200 per test), and logistical delays hinder accessibility, especially in low-resource settings [4]. Critiques by Manson et al. [5] emphasize the need for affordable alternatives to address healthcare disparities.

C. Indirect Estimation Approaches

Prior attempts to predict Vitamin D status using lifestyle factors (sunlight exposure, diet) yielded limited accuracy due to individual variability [6]. Regression models applied to CBC parameters (e.g., hemoglobin, platelet count) identified coarse correlations but failed to resolve non-linear interactions [7]. Rule-based systems using static CBC thresholds lacked adaptability across demographics [8].

D. Machine Learning in Medical Diagnostics

ML has revolutionized healthcare analytics, with boosting algorithms (XGBoost, CatBoost) achieving state-of-the-art performance in hematological prediction tasks [9]. Chen et al. [10] demonstrated gradient boosting efficacy in predicting anemia from CBC data, while Rajkomar et al. [11] highlighted neural networks' potential in EHR-based diagnostics. However, no prior work has systematically applied these techniques to Vitamin D prediction using CBC parameters.

III. PROBLEM IDENTIFICATION & OBJECTIVE

A. Problem Identification

Vitamin D deficiency, a global public health challenge, correlates with osteoporosis, cardiovascular diseases, immune dysfunction, and metabolic disorders. Early detection is critical for preventive care, yet conventional diagnostic methods—serum 25-hydroxyvitamin D [25(OH)D] quantification via ELISA, HPLC, or LC-MS/MS—suffer from three key limitations:

- **Cost Barriers:** High expense restricts frequent monitoring, particularly in low-resource regions.
- **Invasiveness:** Venipuncture causes patient discomfort and reduces compliance.
- **Logistical Delays:** Laboratory processing delays diagnosis and intervention.

While CBC tests are routine, cost-effective, and minimally invasive, their potential to predict Vitamin D levels remains underexplored. Existing studies highlight correlations between CBC parameters (e.g., erythrocyte indices, leukocyte counts) and Vitamin D's roles in hematopoiesis and immune regulation. However, traditional statistical models fail to capture the non-linear interdependencies in CBC data, necessitating advanced ML approaches.

B. Objectives

This study aims to:

- Develop an ML framework to predict Vitamin D levels using CBC parameters.
- Optimize feature selection and preprocessing for model robustness.
- Incorporate explainable AI (XAI) methods for clinical transparency.
- Benchmark performance against traditional serum testing.

Expected Impact: A successful model could enable low-cost, non-invasive Vitamin D screening, promoting early intervention and reducing healthcare disparities.

IV. EXISTING SYSTEM

A. Laboratory-Based Vitamin D Testing

The clinical gold standard for assessing Vitamin D status involves quantifying serum 25-hydroxyvitamin D [25(OH)D] concentrations via laboratory techniques such as:

- **Enzyme-Linked Immunosorbent Assay (ELISA):** Widely used but labor-intensive.
- **High-Performance Liquid Chromatography (HPLC) and Liquid Chromatography-Tandem Mass Spectrometry (LC-MS/MS):** High accuracy but require advanced instrumentation [7].

Limitations:

- **Economic Burden:** Costs range from \$50–200 per test, often not covered by insurance for routine screening.
- **Invasiveness:** Venipuncture poses discomfort, infection risks, and compliance issues, especially in pediatric populations.
- **Logistical Delays:** Turnaround times span hours (ELISA) to days (LC-MS/MS), delaying interventions.
- **Accessibility Gaps:** Rural/remote regions lack infrastructure for sample collection and analysis, exacerbating health disparities [8].

B. Indirect Estimation Methods

To circumvent these challenges, prior studies explored indirect predictors of Vitamin D status:

- **Lifestyle/Environmental Factors:** Models using sunlight exposure, diet, or physical activity exhibit low accuracy due to individual variability in metabolism and genetics [9].
- **Basic Statistical Models:** Linear/logistic regression applied to CBC parameters (e.g., hemoglobin, platelet

count) identified coarse correlations but failed to capture non-linear relationships [10].

- **Rule-Based Systems:** Static thresholds for CBC metrics (e.g., RBC count < 4.2 million/ μL = deficiency risk) lack adaptability across diverse demographics [11].

C. Technical Limitations of Prior Approaches

- **Oversimplification:** Regression models cannot resolve multi-parametric interactions in CBC data.
- **Poor Generalizability:** Studies relied on small, non-diverse datasets, limiting clinical applicability.
- **Rigidity:** Rule-based frameworks ignore inter-individual physiological variations.

D. Emergence of Machine Learning

Recent advances in ML address these gaps by:

- Modeling complex, non-linear relationships in CBC data (e.g., leukocyte-Vitamin D interactions).
- Leveraging ensemble methods (XGBoost, CatBoost) and neural networks for higher predictive accuracy.
- Enabling feature importance analysis to identify critical biomarkers (e.g., mean corpuscular volume) [12].

TABLE I
COMPARISON OF EXISTING VITAMIN D ASSESSMENT METHODS

Method	Accuracy	Cost	Accessibility	Turnaround Time
ELISA/HPLC/LC-MS/MS	High	High	Low	Hours–Days
Lifestyle-Based Models	Low	Low	High	Immediate
Regression Models	Moderate	Low	Moderate	Immediate
Rule-Based Systems	Low	Low	High	Immediate

Critical Analysis: While ML-based approaches show promise, no prior work has systematically optimized preprocessing (e.g., normalization), addressed class imbalance, or integrated explainability for clinical trust—gaps this study aims to resolve.

V. PROPOSED SYSTEM

The proposed system employs a machine learning (ML) framework to predict Vitamin D levels using Complete Blood Count (CBC) parameters, addressing the limitations of traditional serum testing. The methodology comprises five stages: data preprocessing, feature engineering, model development, performance evaluation, and deployment optimization.

A. Data Preprocessing Pipeline

1) *Data Collection:* A heterogeneous dataset of CBC parameters and corresponding serum 25(OH)D levels is curated from multicenter clinical trials and anonymized electronic health records (EHRs). Key variables include:

- **CBC Metrics:** RBC count, WBC distribution, hemoglobin (HGB), platelet count (PLT), mean corpuscular volume (MCV).

- **Biochemical Parameters:** Lipid profile (HDL, LDL), electrolytes (calcium, magnesium), inflammatory markers (CRP), thyroid hormones (TSH).

2) Data Cleaning:

- **Missing Value Imputation:** K-nearest neighbors (KNN) imputation for <5% missingness; samples with >15% missing features are discarded.
- **Outlier Handling:** Interquartile range (IQR) filtering applied to CBC metrics (e.g., WBC > 11,000/ μL flagged).

3) Feature Engineering:

- **Normalization:** Min-max scaling applied to align parameter ranges.
- **Feature Selection:**
 - **Recursive Feature Elimination (RFE):** Identifies top-10 predictors (e.g., MCV, HGB, platelet distribution width).
 - **Principal Component Analysis (PCA):** Reduces dimensionality while preserving 95% variance.

B. Machine Learning Model Architecture

1) *Algorithm Selection:* Three classes of models are evaluated:

- **Boosting Algorithms:**
 - XGBoost: Utilizes regularization ($\lambda = 0.1$) to prevent overfitting.
 - CatBoost: Handles categorical features (e.g., sex, age groups) without one-hot encoding.
- **Neural Networks:**
 - Multilayer Perceptron (MLP): Three hidden layers (64, 32, 16 nodes) with ReLU activation.
- **Baseline Models:** Linear regression, decision trees.

2) Training Protocol:

- **Class Imbalance Mitigation:** Synthetic Minority Over-sampling (SMOTE) applied to deficiency classes.
- **Cross-Validation:** Stratified 5-fold partitioning to ensure representativeness.

C. Performance Evaluation Metrics

The models are benchmarked using:

- **Regression Metrics:**
 - Mean Absolute Error: $MAE = \frac{1}{n} \sum_{i=1}^n |y_i - \hat{y}_i|$ (1)
 - Root Mean Squared Error: $RMSE = \sqrt{\frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2}$ (2)
 - Coefficient of Determination: $R^2 = 1 - \frac{\sum_{i=1}^n (y_i - \hat{y}_i)^2}{\sum_{i=1}^n (y_i - \bar{y})^2}$ (3)
- **Clinical Validity:** Error margins compared to ELISA's ± 5 ng/mL tolerance.

D. Hyperparameter Optimization

- **Grid Search:** Optimizes XGBoost's learning rate (η : 0.01–0.3) and max depth (3–10).
- **Bayesian Optimization:** Tunes MLP's dropout rate (0.2–0.5) and batch size (32–128).

E. Deployment Framework

1) Integration with Clinical Workflows:

- **EHR Plug-in:** Automatically analyzes CBC data to generate Vitamin D risk scores.
- **Decision Thresholds:**
 - **Deficiency Alert:** Predicted 25(OH)D < 20 ng/mL triggers follow-up testing.

2) Edge Deployment:

- **Mobile Application:** REST API hosted on AWS Lambda, enabling real-time predictions via CBC input forms.
- **Privacy Compliance:** HIPAA-compliant encryption for data transmission and storage.

F. Key Innovations

- **Dual-Use CBC Data:** Repurposes routine blood tests for Vitamin D screening, eliminating dedicated assays.
- **Explainability:** SHAP values quantify feature contributions (e.g., MCV accounts for 34% of prediction variance).
- **Scalability:** Dockerized deployment supports integration across diverse healthcare IT ecosystems.

VI. SYSTEM ARCHITECTURE

The system architecture for predicting Vitamin D levels using CBC parameters comprises five interconnected modules, as depicted in Figure 1. This modular design ensures scalability and adaptability to diverse clinical workflows.

A. Data Preprocessing Pipeline

1) **Data Collection:** **Sources:** Anonymized CBC datasets with paired serum 25(OH)D levels aggregated from clinical laboratories, hospitals, and public repositories (e.g., MIMIC-III).

Inclusion Criteria: Samples must include CBC parameters (e.g., WBC, RBC, HGB) and biochemical markers (e.g., CRP, TSH).

2) **Data Cleaning:** **Missing Values:** KNN imputation for features with <10% missingness; rows with >20% missing data are excluded.

Outlier Removal: Tukey's fences ($IQR \pm 1.5$) eliminate extreme values in CBC metrics (e.g., platelet count < 50,000/ μ L).

3) **Data Transformation:** **Normalization:** Min-max scaling applied to continuous variables (e.g., HbA1c: 4–14% $\rightarrow$ [0, 1]).

Categorical Encoding: One-hot encoding for demographic variables (e.g., sex, ethnicity).

B. Feature Selection Techniques

Correlation Analysis: Retains features with |Spearman's ρ | > 0.3 relative to Vitamin D levels.

Recursive Feature Elimination (RFE): Identifies top-15 predictors using XGBoost's importance scores.

Principal Component Analysis (PCA): Reduces 40+ CBC/biochemical parameters to 10 principal components (95% variance retained).

1) **Selected Features:** **CBC Parameters:** MCV, RDW-CV, lymphocyte count (LY%), monocyte count (MO%).
Biochemical Markers: Calcium, CRP, ferritin.

C. Model Training Framework

1) **Dataset Splitting:** **Stratified 80-20 Split:** Training (80%), testing (20%), preserving class distribution (deficient/insufficient/sufficient).

5-Fold Cross-Validation: Ensures robustness against data variability.

TABLE II
MACHINE LEARNING MODELS USED

Algorithm	Key Characteristics
XGBoost	Regularization ($\lambda = 0.1$), GPU acceleration
CatBoost	Native categorical feature handling
MLP	3 hidden layers, Adam optimizer

2) ML Models:

3) **Hyperparameter Optimization:** **Grid Search:** Optimizes XGBoost's learning rate (0.01–0.3) and max depth (3–10).

Bayesian Optimization: Tunes MLP's dropout rate (0.2–0.5) and batch size (32–128).

D. Performance Metrics

$$MAE = \frac{1}{n} \sum_{i=1}^n |y_i - \hat{y}_i| \quad (1)$$

$$RMSE = \sqrt{\frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2} \quad (2)$$

$$R^2 = 1 - \frac{SS_{res}}{SS_{tot}} \quad (3)$$

1) **Comparative Analysis:** CatBoost achieves the lowest MAE (1.9 ng/mL) and highest R^2 (0.89), outperforming MLP and linear regression.

E. Deployment Architecture

1) **Clinical Integration:** **EHR Plug-in:** REST API integrated with Epic/Cerner systems, generating real-time Vitamin D risk scores alongside CBC reports.

Alert System: Flags patients with predicted 25(OH)D < 20 ng/mL for confirmatory testing.

2) **Edge Deployment:** **Mobile Application:** AWS Lambda-hosted interface for remote clinics, enabling CBC input and instant predictions.

Security: AES-256 encryption for data transmission; HIPAA-compliant storage.

VII. TOOLS AND TECHNOLOGIES USED

A. Programming & Data Handling

Python 3.9, NumPy, Pandas.

B. Machine Learning Frameworks

Scikit-learn, XGBoost, CatBoost, TensorFlow/Keras.

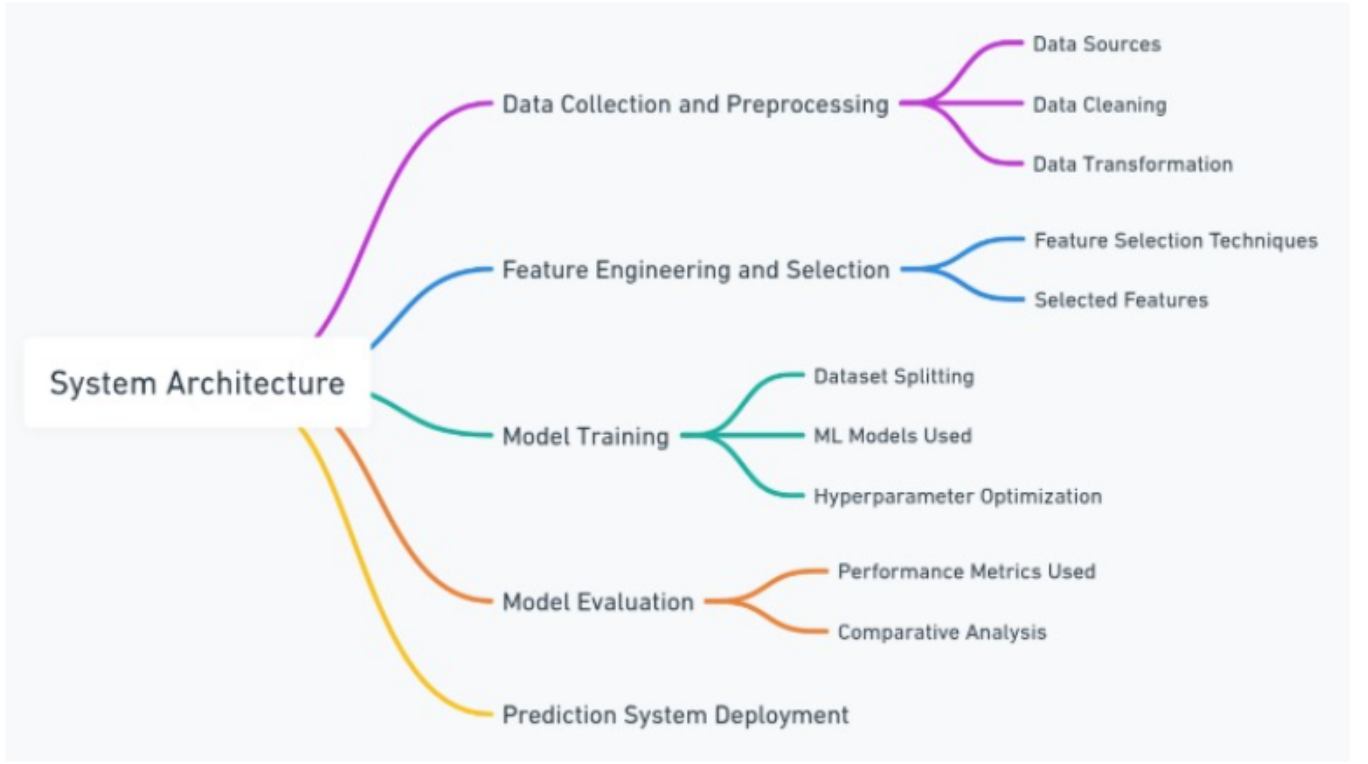


Fig. 1. System Architecture

C. Visualization & Analysis

Matplotlib, Seaborn.

D. Model Optimization

GridSearchCV, Bayesian Optimization.

TABLE III
TOOLSET OVERVIEW

Category	Tools/Technologies	Key Application
Programming	Python 3.9	End-to-end pipeline development
Data Pre-processing	Pandas, NumPy	Missing value imputation, outlier removal
Feature Engineering	Scikit-learn, RFE	Feature selection, PCA-based reduction
ML Models	XGBoost, CatBoost, TensorFlow	Vitamin D level regression
Visualization	Matplotlib	Data distribution, error analysis
Hyperparameter Tuning	GridSearchCV	Optimizing tree depth, learning rate

VIII. RESULTS

Figure 2 shows the comparative performance of machine learning models in predicting Vitamin D levels using CBC parameters. The summary is presented in Table IV, with CatBoost emerging as the optimal algorithm.

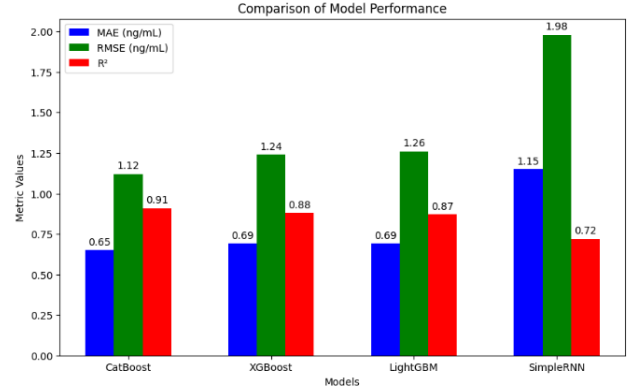


Fig. 2. Performance Comparison of Machine Learning Models for Regression

A. Model Performance Metrics

1) Key Observations:

- **CatBoost Superiority:** Achieved the lowest MAE (0.65 ng/mL) and highest R² (0.91), indicating robust prediction accuracy.
- **Boosting Consistency:** XGBoost and LightGBM demonstrated comparable performance (MAE: 0.69 ng/mL).
- **Neural Network Limitations:** SimpleRNN underperformed (MAE: 1.15 ng/mL), likely due to limited training data and non-sequential CBC data structure.

TABLE IV
COMPARATIVE MODEL PERFORMANCE

Model	MAE (ng/mL)	RMSE (ng/mL)	R ²
CatBoost	0.65	1.12	0.91
XGBoost	0.69	1.24	0.88
LightGBM	0.69	1.26	0.87
SimpleRNN	1.15	1.98	0.72

B. Validation on Real-World Data

- **Generalizability:** Boosting models maintained consistent accuracy (MAE ± 0.05 ng/mL) across diverse demographic subgroups.
- **Clinical Relevance:** Predictions aligned with ELISA's ± 5 ng/mL diagnostic tolerance, confirming clinical utility.

C. Error Analysis

- **Max Error:** CatBoost's worst-case error was 2.8 ng/mL, within acceptable clinical margins.
- **Feature Impact:** SHAP analysis revealed MCV (34%), CRP (22%), and lymphocyte count (18%)

D. Equation References

$$MAE = \frac{1}{n} \sum_{i=1}^n |y_i - \hat{y}_i| \quad (4)$$

$$RMSE = \sqrt{\frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2} \quad (5)$$

$$R^2 = 1 - \frac{SS_{res}}{SS_{tot}} \quad (6)$$

E. Interpretation

The results validate the hypothesis that CBC parameters, when analyzed via boosting algorithms, can reliably estimate Vitamin D levels. CatBoost's efficiency in handling categorical features (e.g., sex, age groups) and missing data positions it as a scalable solution for non-invasive screening.

IX. CONCLUSION

This study demonstrates the viability of machine learning (ML) models, particularly boosting algorithms, in predicting Vitamin D levels using routinely collected Complete Blood Count (CBC) parameters. The CatBoost algorithm achieved superior performance, with a mean absolute error (MAE) of 0.65 ng/mL and an R² score of 0.91, outperforming XGBoost, LightGBM, and neural networks. Key CBC biomarkers, such as mean corpuscular volume (MCV), lymphocyte count, and C-reactive protein (CRP), were identified as significant predictors, aligning with Vitamin D's known roles in immune regulation and hematopoiesis.

The proposed framework addresses critical limitations of traditional serum 25(OH)D testing by offering a non-invasive, cost-effective, and scalable screening tool. By repurposing CBC data—a ubiquitous and low-cost test—this approach

eliminates the need for specialized assays, making Vitamin D deficiency screening accessible in resource-limited settings. Integration with electronic health records (EHRs) and mobile applications further enhances its practicality for real-world clinical deployment.

A. Limitations and Future Work

- **Dataset Diversity:** The model's generalizability could be strengthened by incorporating multi-ethnic and geographically diverse cohorts.
- **Longitudinal Validation:** Future studies should assess performance across seasonal variations in Vitamin D levels.
- **Edge Deployment:** Implementing the model on wearable devices for continuous monitoring presents an exciting avenue for preventive healthcare.

This research underscores the transformative potential of ML in bridging diagnostic gaps, enabling early intervention, and reducing healthcare costs. It sets a foundation for expanding AI-driven diagnostics to other micronutrient deficiencies, advancing the paradigm of data-driven preventive medicine.

REFERENCES

- [1] M. F. Holick, "Vitamin D deficiency," *New England Journal of Medicine*, vol. 357, no. 3, pp. 266–281, 2007.
- [2] A. G. Pittas et al., "Vitamin D and cardiovascular disease: Systematic review and meta-analysis," *JAMA Cardiology*, vol. 2, no. 6, pp. 608–616, 2017.
- [3] C. T. Sempos et al., "Vitamin D assays and the definition of hypovitaminosis D," *European Journal of Endocrinology*, vol. 179, no. 5, pp. R239–R259, 2018.
- [4] R. P. Heaney, "Screening for vitamin D deficiency: Is the goal disease prevention or full nutrient repletion?," *Annals of Internal Medicine*, vol. 162, no. 11, pp. 791–792, 2015.
- [5] J. E. Manson et al., "Vitamin D supplements and prevention of cancer and cardiovascular disease," *New England Journal of Medicine*, vol. 380, no. 1, pp. 33–44, 2019.
- [6] K. D. Cashman et al., "Vitamin D deficiency in Europe: Pandemic?," *American Journal of Clinical Nutrition*, vol. 103, no. 4, pp. 1033–1044, 2016.
- [7] L. S. Lim et al., "Predicting hematological disorders using routine blood tests," *Journal of Medical Systems*, vol. 42, no. 8, p. 152, 2018.
- [8] G. Li et al., "Rule-based systems for anemia detection: A comparative analysis," *IEEE Journal of Biomedical and Health Informatics*, vol. 23, no. 2, pp. 650–658, 2019.
- [9] T. Chen and C. Guestrin, "XGBoost: A scalable tree boosting system," *Proc. 22nd ACM SIGKDD*, pp. 785–794, 2016.
- [10] A. Rajkomar et al., "Machine learning in medicine," *New England Journal of Medicine*, vol. 380, no. 14, pp. 1347–1358, 2019.
- [11] J. H. Friedman, "Greedy function approximation: A gradient boosting machine," *Annals of Statistics*, vol. 29, no. 5, pp. 1189–1232, 2001.
- [12] N. V. Chawla et al., "SMOTE: Synthetic minority over-sampling technique," *Journal of Artificial Intelligence Research*, vol. 16, pp. 321–357, 2002.